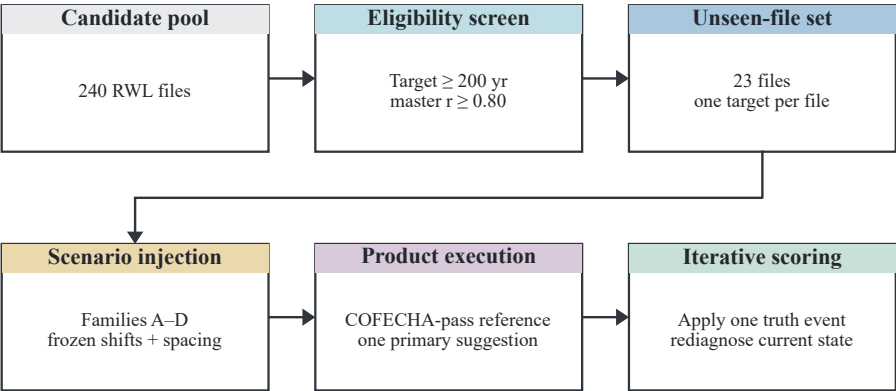


Cross-file benchmark design and capability coverage

23 unseen RWL files • four scenario families • file-cluster uncertainty • product-identical execution

a File-disjoint validation pipeline



b Four families span increasing event complexity

Family	Structure	Scen.	Cases	Truths	Capability
A	Clean + single events	5	115	92	Core operation
B	Distant pairs / whole + local	12	276	552	Separated recovery
C	Near local pairs (7 yr)	9	207	414	Local discrimination
D	Whole + two local events	4	92	276	Global + local

c Scenario coverage across diagnostic capabilities

	Clean control	Unit edit	Range shift	Whole shift	Near events	Mixed events
A Single	YES	YES	YES	YES		
B Distant		YES	YES	YES		YES
C Near		YES	YES		YES	YES
D Mixed		YES	YES	YES	YES	YES

d Reproducibility controls and update path

